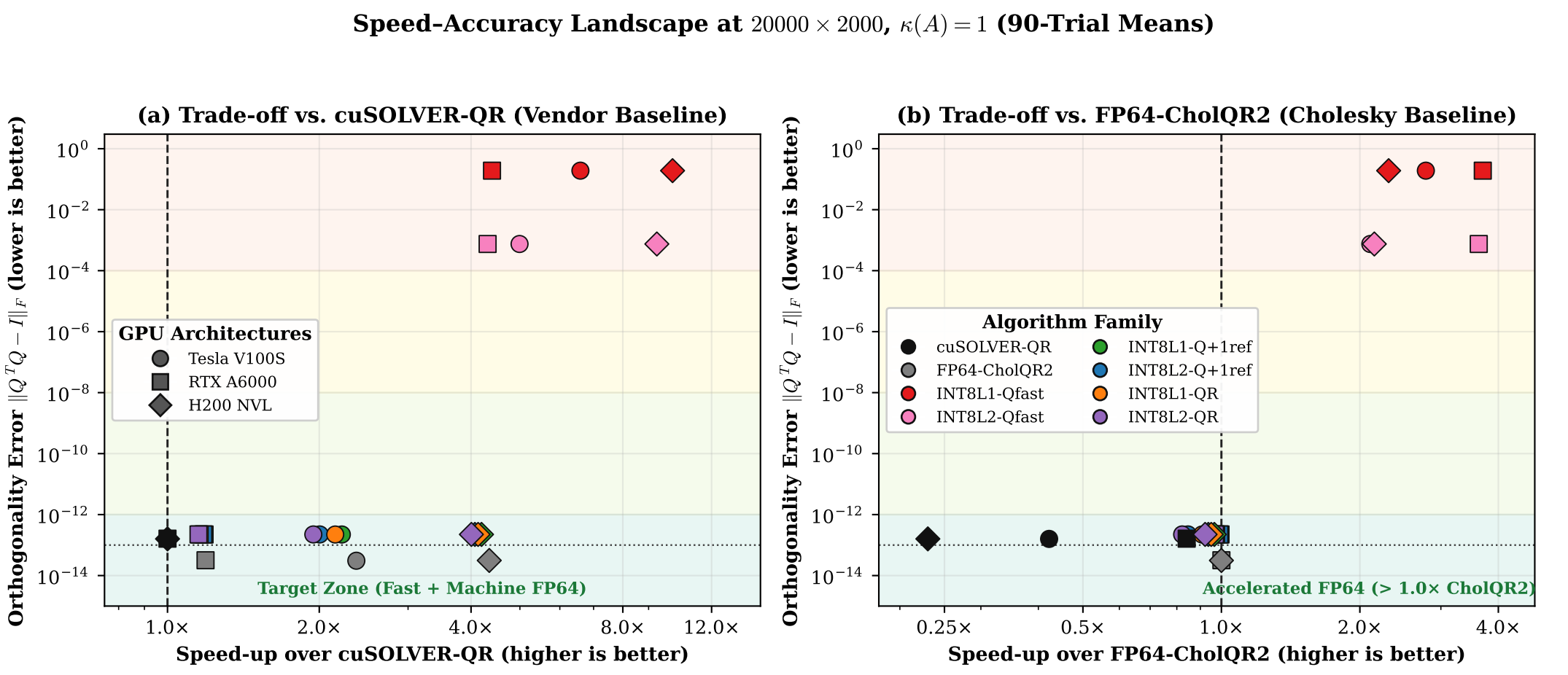
Speed-Accuracy Landscape at 20000×2000, $\kappa(A) = 1$ (90-Trial Means)
(a) Trade-off vs. cuSOLVER-QR (Vendor Baseline)
(b) Trade-off vs. FP64-CholQR2 (Cholesky Baseline)
Orthogonality Error $\|Q^TQ - I\|_F$ (lower is better)
Speed-up over cuSOLVER-QR (higher is better)
Speed-up over FP64-CholQR2 (higher is better)
GPU Architectures
Tesla V100S
RTX A6000
H200 NVL
Algorithm Family
cuSOLVER-QR
FP64-CholQR2
INT8L1-Qfast
INT8L2-Qfast
INT8L1-Q+1ref
INT8L2-Q+1ref
INT8L1-QR
INT8L2-QR
Target Zone (Fast + Machine FP64)
Accelerated FP64 (> 1.0× CholQR2)